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Scaled Sparse Linear Regression

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Abstract

Scaled sparse linear regression jointly estimates the regression coefficients and noise level in a linear model. It chooses an equilibrium with a sparse regression method by iteratively estimating the noise level via the mean residual squares and scaling the penalty in proportion to the estimated noise level. The iterative algorithm costs nearly nothing beyond the computation of a path of the sparse regression estimator for penalty levels above a threshold. For the scaled Lasso, the algorithm is a gradient descent in a convex minimization of a penalized joint loss function for the regression coefficients and noise level. Under mild regularity conditions, we prove that the method yields simultaneously an estimator for the noise level and an estimated coefficient vector in the Lasso path satisfying certain oracle inequalities for the estimation of the noise level, prediction, and the estimation of regression coefficients. These oracle inequalities provide sufficient conditions for the consistency and asymptotic normality of the estimator for the noise level, including cases where the number of variables is of greater order than the sample size. Numerical results demonstrate the superior performance of the proposed method over an earlier proposal of joint convex minimization.

Keywords: Variance estimation, convex minimization, oracle inequality, consistency, asymptotic normality, convergence in distribution, linear regression, penalized least squares, iterative algorithm.

Running title: Scaled Lasso

1. Introduction. This paper concerns the simultaneous estimation of the regression coefficients and noise level in a high-dimensional linear model. High-dimensional data analysis is a topic of great current interest in statistics due to the growth of applications where the number of unknowns far exceeds the number of data points. Among such problems, linear regression is the best understood. Penalization, convex minimization and thresholding methods have been proposed, tested with real and simulated data, and proved to control errors in prediction, estimation and variable

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selection under various sets of regularity conditions. These methods typically require a proper penalty or threshold level to achieve their goals. A larger penalty level may lead to a simple model with large bias, while a smaller penalty level may lead to a complex noisy model due to over fitting. Scale-invariance considerations and existing theory suggest that the penalty level should be proportional to the noise level of the regression model. In the absence of the knowledge of the noise level, cross-validation is commonly used to determine the penalty level. However, cross-validation is computationally costly and theoretically poorly understood, especially for the purpose of variable selection and the estimation of regression coefficients.

Estimation of the noise level in high-dimensional regression is an interesting problem on its own right. Examples include quality control in manufacturing and volatility control in finance.

Our study is motivated by Städler, Bühlmann and van de Geer (2010) and the comments on that paper by Antoniadis (2010) and Sun and Zhang (2010). Städler, Bühlmann and van de Geer (2010) proposed to estimate the regression coefficients and noise level by maximizing their joint log-likelihood with an ℓ_1 penalty on the regression coefficients. Their method has a unique solution due to the joint concavity of the log-likelihood under a certain transformation of the unknown parameters. Sun and Zhang (2010) proved that this penalized joint MLE may result in a positive bias for the estimation of the noise level and compared it with two alternatives. The first alternative is a one-step bias correction of the penalized joint MLE. The second alternative is an iterative algorithm which alternates between estimating the noise level via the mean residual squares and scaling the penalty level in a predetermined proportion to the estimated noise level in the Lasso or MC+ paths. Sun and Zhang (2010) demonstrated in a simulation experiment the superiority of the iterative algorithm for both the Lasso and MC+, compared with the penalized joint MLE and its bias correction. However, no theoretical results were given for the convergence or consistency of the iterative algorithm. Antoniadis (2010) commented on the same problem from a different perspective by raising the possibility of adding an ℓ_1 penalty to Huber's (1981) joint convex loss function. Interestingly, the minimizer of this penalized joint convex loss is identical to the equilibrium of the iterative algorithm for the Lasso path. Thus, the convergence of the iterative algorithm is guaranteed by the convexity.

In this paper, we study Sun and Zhang's (2010) iterative algorithm with the Lasso and MC+ paths for the joint estimation of regression coefficients and the noise level. For the Lasso, this is equivalent to minimizing Huber's (1981) joint loss function with the ℓ_1 penalty as Antoniadis (2010) pointed out. For simplicity in later references, we call the equilibrium of this algorithm as the scaled version of the penalized regression method, for example scaled Lasso or scaled MC+, depending on the choice of the penalty function. Under mild regularity conditions, we prove certain oracle inequalities for prediction and the joint estimation of the noise level and regression coefficients for the scaled Lasso. These oracle inequalities imply the consistency and

asymptotic normality of the scaled Lasso estimator for the noise level. We report numerical results to demonstrate the performance of scaled Lasso and MC+. These theoretical and numerical results strongly support the use of the proposed method for high-dimensional regression.

The rest of the paper is organized as follows. In Section 2, we describe the iterative algorithm and its connection to convex minimization. In Section 3, we provide oracle inequalities for the scaled Lasso and prove the consistency and asymptotic normality of the estimator for the noise level. In Section 4, we present numerical results. Section 5 contains some discussion, including oracle inequalities for the Lasso with a predetermined penalty level. An appendix provides all proofs.

We use the following notation throughout the paper. For a vector $v = (v_1, \dots, v_p)$, $|v|_q = (\sum_j |v_j|^q)^{1/q}$ is the ℓ_q norm with the usual extension $|v|_\infty = \max_j |v_j|$ and $|v|_0 = \#\{j : v_j \neq 0\}$. Let X denote a design matrix and for subsets A of $\{1, \dots, p\}$, $X_A = (x_j, j \in A)$. Moreover, $x_+ = \max(x, 0)$.

2. An iterative algorithm. In this section, we describe the iterative algorithm for the joint estimation of regression coefficients and noise level and its connection to convex minimization.

Suppose we observe a design matrix $X = (x_1, \dots, x_p) \in \mathbb{R}^{n \times p}$ and a response vector $y \in \mathbb{R}^n$. For penalty functions $\rho(\cdot)$, consider penalized loss functions of the form

$$L_\lambda(\beta) = \frac{|y - X\beta|_2^2}{2n} + \lambda^2 \sum_{j=1}^p \rho(|\beta_j|/\lambda) \quad (1)$$

where $\beta = (\beta_1, \dots, \beta_p)'$ is a vector of regression coefficients. Let the penalty $\rho(t)$ be standardized to $\dot{\rho}(0+) = 1$, where $\dot{\rho}(t) = (d/dt)\rho(t)$. A vector $\hat{\beta} = (\hat{\beta}_1, \dots, \hat{\beta}_p)'$ is a critical point of the penalized loss (1) if and only if

$$\begin{cases} x'_j(y - X\hat{\beta})/n = \lambda \text{sgn}(\hat{\beta}_j) \dot{\rho}(|\hat{\beta}_j|/\lambda), & \hat{\beta}_j \neq 0 \\ x'_j(y - X\hat{\beta})/n \in \lambda[-1, 1], & \hat{\beta}_j = 0. \end{cases} \quad (2)$$

If the penalized loss (1) is convex in β , (2) is the Karush-Kuhn-Tucker (KKT) conditions for its minimization. For quadratic spline penalties $\rho(t)$ with m knots, Zhang (2007, 2010) developed a PLUS algorithm to compute a linear spline path of solutions $\{\lambda^{(t)} \oplus \hat{\beta}^{(t)}, t \geq 0\}$ of (2) to cover the entire range of λ . This extends the LARS/Lasso path (Osborne *et al.*, 2000a, 2000b; Efron *et al.*, 2004) from $m = 1$ and includes the minimax concave penalty (MCP) for $m = 2$ and the SCAD penalty (Fan and Li, 2001) for $m = 3$. An R package “plus” is available for computing the solution paths for these penalties.

Given a solution path $\hat{\beta}(\lambda)$ satisfying (2), one still has to choose a penalty level λ to arrive at an estimator of the unknown β . Such a choice may depend on the purpose of estimation, since variable selection may require a larger λ than prediction. However,

scale-invariance consideration and existing theoretical results suggest a penalty level in proportion to the noise level σ . This motivates the iterative algorithm

$$\begin{cases} \hat{\sigma}^2 \leftarrow |y - X\hat{\beta}|_2^2 / \{(1-a)n\} \\ \lambda \leftarrow \hat{\sigma}\lambda_0 \\ \hat{\beta} \leftarrow \hat{\beta}(\lambda) \end{cases} \quad (3)$$

in the solution path, where $\hat{\sigma}$ can be viewed as an estimate for the noise level and $a \geq 0$ provides an option for a degrees-of-freedom adjustment with $a > 0$. For $p < n$ and $(a, \lambda_0) = (p/n, 0)$, (3) is non-iterative and gives the least squares estimator $\hat{\beta}_{LSE}$ and $\hat{\sigma}^2 = |y - X\hat{\beta}_{LSE}|_2^2 / (n - p)$. For scaled Lasso, we use $a = 0$ in (3) and $\rho(t) = t$ in (1) and (2). For scaled MCP+, we use $a = 0$ and the MCP $\rho(t) = \int_0^t (1 - x/\gamma)_+ dx$, where $\gamma > 0$ regularizes the maximum concavity of the penalty. The scaled MCP+ with $\gamma = \infty$ becomes the scaled Lasso. The algorithm (3) can be easily implemented once a solution path is computed.

Consider the ℓ_1 penalty. As discussed in the introduction, (3) is an alternating minimization algorithm for the penalized joint loss function

$$L_{\lambda_0}(\beta, \sigma) = \frac{|y - X\beta|_2^2}{2n\sigma} + \frac{(1-a)\sigma}{2} + \lambda_0|\beta|_1. \quad (4)$$

For $a = 0$ and $\lambda = \hat{\sigma}\lambda_0$ with fixed $\hat{\sigma}$, $\hat{\sigma}L_{\lambda_0}(\beta, \hat{\sigma}) = L_{\lambda}(\beta) + \hat{\sigma}^2/2$, so that $\hat{\beta} \leftarrow \hat{\beta}(\lambda)$ in (3) minimizes $L_{\lambda_0}(\beta, \hat{\sigma})$ over β . For fixed $\hat{\beta}$, $\hat{\sigma}^2 \leftarrow |y - X\hat{\beta}|_2^2 / \{(1-a)n\}$ in (3) minimizes $L_{\lambda_0}(\hat{\beta}, \sigma)$ over σ . We summarize some properties of the algorithm (3) in the following proposition.

Proposition 1 *Let $\hat{\beta} = \hat{\beta}(\lambda)$ be a solution path of (2) with $\rho(t) = t$. The penalized loss function (4) is jointly convex in $\{\beta, \sigma\}$ and the algorithm (3) converges to*

$$\{\hat{\beta}, \hat{\sigma}\} = \arg \min_{\beta, \sigma} L_{\lambda_0}(\beta, \sigma). \quad (5)$$

The resulting estimator $\hat{\beta} = \hat{\beta}(X, y)$ and $\hat{\sigma} = \hat{\sigma}(X, y)$ are scale equivariant in y in the sense that $\hat{\beta}(X, cy) = c\hat{\beta}(X, y)$ and $\hat{\sigma}(X, cy) = |c|\hat{\sigma}(X, y)$. Moreover,

$$\frac{\partial}{\partial \sigma} L_{\lambda_0}(\hat{\beta}(\sigma\lambda_0), \sigma) = \frac{1-a}{2} - \frac{|y - X\hat{\beta}(\sigma\lambda_0)|_2^2}{2n\sigma^2}. \quad (6)$$

3. Theoretical results. In this section, we study theoretical properties of the scaled Lasso (3). Let β^* be a vector of true regression coefficients. An oracle expert with the knowledge of β^* would estimate the noise level by $\sigma^* = |y - X\beta^*|_2 / \sqrt{n}$. We derive upper and lower bounds for $\hat{\sigma}/\sigma^* - 1$ and use them to prove the consistency and asymptotic normality of $\hat{\sigma}$. We derive oracle inequalities for the prediction performance and the estimation of β under the ℓ_q loss. Throughout the sequel, $P_{\beta, \sigma}$ is

the probability measures under which $y - X\beta \sim N(0, \sigma^2 I_n)$. We assume $|x_j|_2^2 = n$ whenever $P_{\beta, \sigma}$ is invoked.

We first provide the consistency for the estimation of σ via an oracle inequality for the persistency (Greenshtein and Ritov, 2004) of the scaled Lasso. For $\lambda > 0$, $\xi > 1$, $w \in \mathbb{R}^p$, and $T \subset \{1, \dots, p\}$, define

$$\eta(\lambda, \xi, w, T) = |X\beta^* - Xw|_2^2/n + 4\lambda|w_{T^c}|_1 + \frac{4\xi^2\lambda^2|T|}{(\xi + 1)^2\kappa^2(\xi, T)} \quad (7)$$

where $\kappa(\xi, T)$, the compatibility constant (van de Geer, 2007), is defined as

$$\kappa(\xi, T) = \min \left\{ \frac{|T|^{1/2}|Xu|_2}{n^{1/2}|u_T|_1} : u \in \mathcal{C}(\xi, T), u \neq 0 \right\} \quad (8)$$

with the cone $\mathcal{C}(\xi, T) = \{u : |u_{T^c}|_1 \leq \xi|u_T|_1\}$. Define

$$\eta_*(\lambda, \xi) = \inf_{w, T} \eta(\lambda, \xi, w, T), \quad \tau_0 = \eta_*^{1/2}(\sigma^*\lambda_0, \xi)/\sigma^*. \quad (9)$$

Theorem 1 *Let $\{\hat{\beta}, \hat{\sigma}\}$ be as in (5), $\beta^* \in \mathbb{R}^p$, $\sigma^* = |y - X\beta^*|_2/\sqrt{n}$, $z^* = |X'(y - X\beta^*)/n|_\infty/\sigma^*$ and $\xi > 1$. In the event $z^* \leq (1 - \tau_0)\lambda_0(\xi - 1)/(\xi + 1)$,*

$$-\tau_0 \leq \frac{\hat{\sigma}}{\sigma^*} - 1 \leq \frac{|X\hat{\beta} - X\beta^*|_2}{\sqrt{n}\sigma^*} \leq \frac{1}{\sigma^*} \eta_*^{1/2}\left(\frac{\sigma^*\lambda_0}{1 - \tau_0}, \xi\right) \leq \frac{\tau_0}{1 - \tau_0}. \quad (10)$$

In particular, if $\lambda_0 = A\sqrt{(2/n)\log p}$ with $A > (\xi + 1)/(\xi - 1)$, then

$$\eta_*(\sigma\lambda_0, \xi)/\sigma \rightarrow 0 \Rightarrow P_{\beta^*, \sigma}(|\hat{\sigma}/\sigma - 1| > \epsilon) \rightarrow 0 \quad \forall \epsilon > 0. \quad (11)$$

Theorem 1 extends to the scaled Lasso a unification of prediction oracle inequalities for a fixed penalty. For fixed penalty, the error bound in van de Geer and Bühlmann (2009) requires $w = \beta^*$ and $\beta_{T^c}^* = 0$ in (7) and those in Koltchinskii, Tsybakov and Lounici (2010) require either $w_{T^c} = 0$ or $T = \emptyset$, whereas the minimum is taken over all $\{w, T\}$ in (9). This will be further discussed in Section 5. A consequence of this persistency of the scaled Lasso is the consistency of the corresponding estimator of the noise level in (11). Due to the scale equivariance in Proposition 1, Theorem 1 and the results in the rest of the section are all scale free.

Now we provide sharper convergence rates and the asymptotic normality for the scaled Lasso estimation of the noise level σ . This sharper rate is based on the following ℓ_1 error bound for the estimation of β :

$$\mu(\lambda, \xi) = (\xi + 1) \min_T \inf_{0 < \nu < 1} \max \left[\frac{|\beta_{T^c}^*|_1}{\nu}, \frac{\lambda|T|/\{2(1 - \nu)\}}{\kappa^2((\xi + \nu)/(1 - \nu), T)} \right]. \quad (12)$$

Theorem 2 Let $\{\widehat{\beta}, \widehat{\sigma}, \beta^*, \sigma^*, z^*, \xi\}$ be as in Theorem 1. Set

$$\tau(\sigma) = \lambda_0 \mu(\sigma \lambda_0, \xi) / \sigma, \quad \tau_+ = \frac{\tau(\sigma^*) \xi}{(\xi + 1) \{1 - \tau(\sigma^*)\}_+}, \quad \tau_- = \frac{\tau(\sigma^*) (\xi - 1)}{\xi + 1}. \quad (13)$$

(i) The following inequalities hold in the event $z^* \leq (1 - \tau_-) \lambda_0 (\xi - 1) / (\xi + 1)$,

$$-\tau_- \leq \widehat{\sigma} / \sigma^* - 1 \leq \tau_+, \quad |\widehat{\beta} - \beta^*|_1 \leq \mu(\sigma^* \lambda_0, \xi) (1 + \tau_+). \quad (14)$$

(ii) Let $\lambda_0 \geq \{(2/n) \log(p/\epsilon)\}^{1/2} (\xi + 1) / \{(\xi - 1)(1 - \tau_-)\}$. For $\log(p/\epsilon) < n - 2$, $z^* \leq (1 - \tau_-) \lambda_0 (\xi - 1) / (\xi + 1)$ happens with at least probability $1 - (1 + o(1)) \epsilon / \sqrt{\pi \log(p/\epsilon)}$ under $P_{\beta^*, \sigma}$. Moreover, if $\lambda_0 = A \sqrt{(2/n) \log p}$ with $A > (\xi + 1) / (\xi - 1)$, then

$$\tau(\sigma) = o(n^{-1/2}) \Rightarrow \sqrt{n} \left(\frac{\widehat{\sigma}}{\sigma} - 1 \right) \xrightarrow{D} N(0, 1/2) \quad \text{under } P_{\beta^*, \sigma}. \quad (15)$$

Since $\tau(\sigma) \lesssim \min_T \eta(\sigma \lambda_0, 2\xi, \beta^*, T) / \sigma^2$, (14) is of a sharper form than (10) with the square-root for τ_0 in (9). This leads to the asymptotic normality (15). Thus, statistical inference about σ is justified with the scaled Lasso in certain “large p smaller n ” cases, for example, when $\beta_{S^c}^* = 0$ and $|S|(\log p)/n \rightarrow 0$ under the compatibility condition (van de Geer, 2007).

As in Zhang and Huang (2008), Zhang (2009), and Ye and Zhang (2010), the ℓ_1 oracle inequality in (14) allows $|\beta^*|_0 = p$ as long as $|\beta_{T^c}^*|$ is small for some T . For $|\beta_{S^c}^*|_1 = 0$, the bound $\mu(\lambda, \xi)$ improves upon the oracle inequality in van de Geer and Bühlmann (2009) by a constant factor. This will be further discussed in Section 5.

For a fixed penalty level, oracle inequalities for the ℓ_q error of the Lasso have been established in Bunea, Tsybakov and Wegkamp (2007), van de Geer (2008), and van de Geer and Bühlmann (2009) for $q = 1$, Zhang and Huang (2008) and Bickel, Ritov and Tsybakov (2009) for $q \in [1, 2]$, Meinshausen and Yu (2009) for $q = 2$, and Zhang (2009) and Ye and Zhang (2010) for $q \geq 1$.

The bounds on $\widehat{\sigma} / \sigma^*$ in (14), with $\max\{\tau_-, \tau_+ / (1 + \tau_+)\} \leq \tau(\sigma^*)$ for $\tau(\sigma^*) \leq 1$, allow automatic extensions of these existing ℓ_q oracle inequalities from the Lasso with fixed penalty to the scaled Lasso. We illustrate this point by extending the oracle inequalities of Ye and Zhang (2010) for the Lasso and Candès and Tao (2007) for the Dantzig selector in the following corollary. Let

$$SCIF_q(\xi, S) = \inf \left\{ \frac{|S|^{1/q} |X' X u|_\infty}{n |u|_q} : u \in \mathcal{C}_-(\xi, S) \right\} \quad (16)$$

be the sign-restricted cone invertibility factor (Ye and Zhang, 2010), where $\mathcal{C}_-(\xi, S) = \{u : |u_{S^c}|_1 \leq \xi |u_S|_1 \neq 0, u_j x_j' X u \leq 0, \forall j \notin S\}$. For $(|A|, |B|, |u|_2) = (\lceil a \rceil, \lceil b \rceil, 1)$ with $A \cap B = \emptyset$, define

$$\delta_a^\pm = \max_{A, u} \left\{ \pm \left(|X_A u|_2 / \sqrt{n} - 1 \right) \right\}, \quad \theta_{a, b} = \max_{A, B, u} |X_A' X_B u|_2 / n. \quad (17)$$

Corollary 1 Suppose $|\beta_{Sc}^*|_1 = 0$. Then, Theorem 2 holds with $\mu(\lambda, \xi)$ replaced by $\lambda|S|(2\xi)/\{(\xi+1)SCIF_1(\xi, S)\}$, and for $z^* \leq (1-\tau_-)\lambda_0(\xi-1)/(\xi+1)$,

$$|\hat{\beta} - \beta^*|_q \leq \frac{k^{1/q}(\sigma^* z^* + \hat{\sigma}\lambda_0)}{SCIF_q(\xi, S)} \leq \frac{2\sigma^*\xi(1+\tau_+)\lambda_0 k^{1/q}}{(\xi+1)SCIF_q(\xi, S)} \quad (18)$$

for all $q \geq 1$, where $k = |S|$. In particular, for $\xi = \sqrt{2}$ and $z^* \leq (1-\tau_-)\lambda_0(\sqrt{2}-1)^2$,

$$|\hat{\beta} - \beta^*|_2 \leq \frac{\sqrt{8k}(1+\tau_+)\lambda_0\sigma^*}{(\sqrt{2}+1)SCIF_2(\sqrt{2}, S)} \leq \frac{4(1+\tau_+)\sqrt{k}\lambda_0\sigma^*}{(\sqrt{2}+1)(1-\delta_{1.5k}^- - \theta_{2k,1.5k})_+}. \quad (19)$$

The proof of Theorem 1 and 2 in this section is based on the basic inequality

$$(Xw - X\hat{\beta}(\lambda))'(X\beta^* - X\hat{\beta}(\lambda))/n \leq \lambda(|w|_1 - |\hat{\beta}(\lambda)|_1) + \sigma^* z^* |w - \hat{\beta}(\lambda)|_1 \quad (20)$$

as an immediate consequence of the KKT conditions (2). This basic inequality is used to derive upper and lower bounds for (6), the derivative of the profile loss function $L_{\lambda_0}(\hat{\beta}(\sigma\lambda_0), \sigma)$ with respect to σ , within a neighborhood of $\sigma/\sigma^* = 1$. The general w is used in Theorem 1, while $w = \beta^*$ is taken in Theorem 2. The bounds for the minimizer $\hat{\sigma}/\sigma^*$ then follow from the joint convexity of the penalized loss (4). The basic inequality (20) also guarantees the membership of $\hat{\beta} - \beta^*$ in the cone $\mathcal{C}_-(\xi, S)$, so that the quantities (8) and (16) can be applied.

4. Numerical results. In this section, we present some numerical results to compare five methods: the scaled Lasso, MC+, and SCAD (Fan and Li, 2001), the ℓ_1 penalized maximum likelihood estimator (PMLE) of Städler, Bühlmann and van de Geer (2010), and the bias corrected PMLE (BC) of Sun and Zhang (2010). The PMLE is

$$\{\hat{\beta}_{PMLE}, \hat{\sigma}_{PMLE}\} = \arg \max_{\beta, \sigma} \left\{ -\frac{|y - X\beta|_2^2}{2\sigma^2 n} - \log \sigma - \lambda_0 \frac{|\beta|_1}{\sigma} \right\},$$

or equivalently the limit of the iteration $\hat{\sigma} \leftarrow \{y'(y - X\hat{\beta})/n\}^{1/2}$ and $\hat{\beta} \leftarrow \hat{\beta}(\hat{\sigma}\lambda_0)$. The BC is one iteration of (3) from PMLE with $a = 0$,

$$\hat{\sigma}_{BC} = |y - \hat{\beta}(\hat{\sigma}_{MLE}\lambda_0)|_2/\sqrt{n}, \quad \hat{\beta}_{BC} = \hat{\beta}(\hat{\sigma}_{BC}\lambda_0).$$

Two simulation examples and a real data set are considered.

4.1. Simulation study.

Example 1. This experiment has the same setting as in the simulation study in Sun and Zhang (2010) where the scaled Lasso and MC+ are called naive Lasso and MC+ respectively. We provide the description of the simulation settings in Sun and Zhang (2010) in our notation as follows: $(n, p) = (600, 3000)$, x_j are normalized columns from a Gaussian random matrix with iid rows and the correlation $\sigma_{j,k} =$

Table 1: Simulation results for Example 1.

Method		$\sigma_{1,2} = 0.1$		$\sigma_{1,2} = 0.9$	
		Bias $\pm$ SE ($\widehat{\sigma}$)	$ \widehat{\beta} - \beta^* _2$	Bias $\pm$ SE ($\widehat{\sigma}$)	$ \widehat{\beta} - \beta^* _2$
PMLE	λ_1	0.546 $\pm$ 0.035	0.871	0.249 $\pm$ 0.027	0.534
	λ_2	0.768 $\pm$ 0.040	1.210	0.381 $\pm$ 0.029	0.542
	λ_3	0.947 $\pm$ 0.040	1.499	0.574 $\pm$ 0.034	0.576
BC	λ_1	0.316 $\pm$ 0.038	0.778	0.037 $\pm$ 0.027	0.537
	λ_2	0.616 $\pm$ 0.046	1.143	0.126 $\pm$ 0.029	0.536
	λ_3	0.908 $\pm$ 0.047	1.489	0.319 $\pm$ 0.035	0.560
Scaled Lasso	λ_1	0.185 $\pm$ 0.038	0.728	0.006 $\pm$ 0.028	0.538
	λ_2	0.506 $\pm$ 0.052	1.091	0.068 $\pm$ 0.029	0.535
	λ_3	0.892 $\pm$ 0.051	1.485	0.190 $\pm$ 0.034	0.553
Scaled MC+	λ_1	-0.016 $\pm$ 0.037	0.468	0.007 $\pm$ 0.027	0.817
	λ_2	0.179 $\pm$ 0.056	0.743	0.066 $\pm$ 0.028	0.780
	λ_3	0.787 $\pm$ 0.081	1.396	0.180 $\pm$ 0.033	0.719
Scaled SCAD	λ_1	0.068 $\pm$ 0.041	0.603	0.009 $\pm$ 0.027	0.765
	λ_2	0.480 $\pm$ 0.061	1.097	0.069 $\pm$ 0.029	0.669
	λ_3	0.892 $\pm$ 0.051	1.485	0.189 $\pm$ 0.034	0.582

$\sigma_{1,2}^{|k-j|}$ among entries within each row, $\gamma = 2/(1 - \max |x'_k x_j|/n)$ for the MC+ and SCAD, the nonzero β_j^* are composed of 5 blocks of $\beta_*(1, 2, 3, 4, 3, 2, 1)'$ centered at random multiples $j_1, \dots, j_5$ of 25, β_* sets $|X\beta^*|_2^2 = 3n$, and $y - X\beta^*$ is a vector of iid $N(0, 1)$ variables ($\sigma = 1$). It has two settings: $\sigma_{1,2} = 0.1$ for low correlation and $\sigma_{1,2} = 0.9$ for high correlation. We compute each estimator at three penalty levels λ_0 , $\lambda_1 = \{(\log p)/n\}^{1/2}$, $\lambda_2 = \{2(\log p)/n\}^{1/2}$ and $\lambda_3 = 2\{(\log p)/n\}^{1/2}$.

We summarize the simulation results in Table 1, which provides the mean and standard deviation of the estimates of σ and the average ℓ_2 error of the estimation of regression coefficients based on 100 replications. In this simulation experiment, the scaled Lasso outperforms the PMLE and BC, the two other estimates based on the Lasso path. Moreover, we observe that scaled MC+ and SCAD perform well at the “universal penalty level” $\lambda_2 = \{(2/n) \log p\}^{1/2}$. The simulation results also suggest that smaller λ may provide a somewhat better estimate of the noise level at the cost of a larger estimation error for the coefficients, possibly caused by selecting an incorrect model.

Since $SCIF(\xi, S) > 1$ and $|S|(\log p)/n > 0.333$, the condition for the asymptotic normality in Theorem 2 holds marginally at the best with an approximate variance $(1200)^{-1/2} = 0.029$. We plot the histogram of the simulated $\widehat{\sigma}$ for the scaled estimators with the penalty level λ_1 in Figure 1, with the approximate normal density super-imposed. The plots demonstrate a reasonable match between the simulations and asymptotic theory, except for the scaled Lasso and SCAD in the case of low

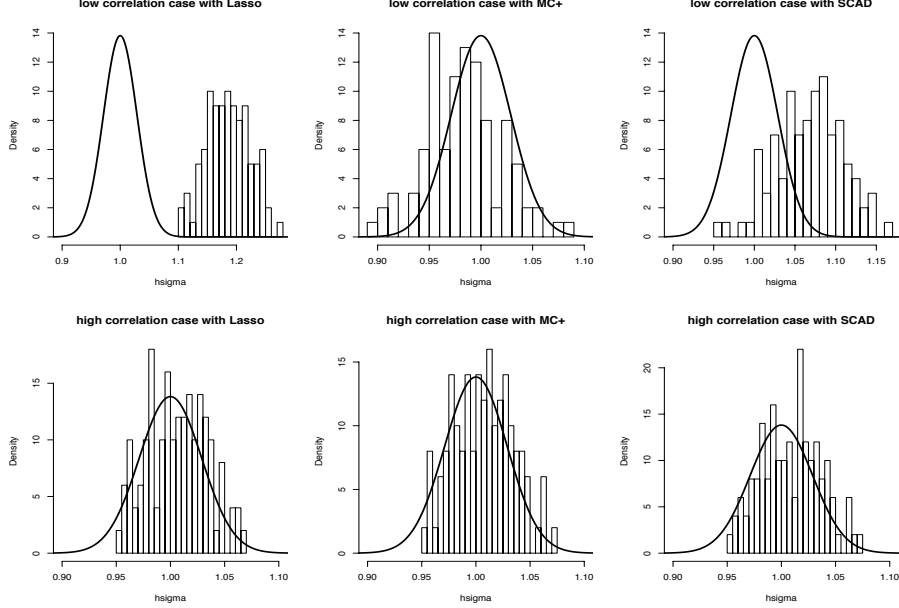


Figure 1: *Top panel: $\sigma_{1,2} = 0.1$; Bottom panel: $\sigma_{1,2} = 0.9$; Left: the scaled Lasso; Middle: the scaled MC+; Right: the scaled SCAD*

correlation. For λ_2 and λ_3 , the bias of the scaled estimators dominates the standard error as show in Table 1.

Example 2. We compare the five estimators at the same three penalty levels $\{\lambda_1, \lambda_2, \lambda_3\}$ as in Example 1. The experiment has the same setting as Example 2 of Fan, Guo and Hao (2010), with the smallest signal ($b = 1/\sqrt{3}$). Fan, Guo and Hao (2010) considered several joint estimators of $\{\beta, \sigma\}$ using cross-validation. Their results are included without repeating their experiment. We provide Fan, Guo and Hao's (2010) description of the simulation setting in our notation as follows. The design matrix X has i.i.d rows and each row follows a multinormal distribution with marginal distribution $N(0, 1)$, $\text{corr}(x_i, x_j) = \sigma_{1,2}, \forall i < j \leq 50$ and $\text{corr}(x_i, x_j) = 0$ otherwise. Let $(n, p) = (200, 2000)$ and nonzero coefficients $\beta_j = 1/\sqrt{3}$ for $j \in S = \{1, 2, 3\}$, and $y - X\beta \sim N(0, \sigma^2 I)$ with $\sigma = 1$. Two configurations are considered: independent columns x_j with $\sigma_{1,2} = 0$ and correlated first 50 columns x_j with $\sigma_{1,2} = 0.5$. Again, we set $\gamma = 2/(1 - \max |x'_k x_j|/n)$ for the MC+ and SCAD.

Our simulation results are presented in the top section of Table 2, while Fan, Guo and Hao's (2010) results are included in the bottom section. Fan, Guo and Hao (2010) proposed a refitted cross-validation (RCV) to estimate σ with the sure independent screening (SIS) and the iterative SIS (ISIS) of Fan and Lv (2008), the SCAD and the LASSO estimators, and compared these with a naive estimator (N)

Table 2: Simulation results for Example 2.

Method		$\sigma_{1,2} = 0$			$\sigma_{1,2} = 0.5$		
		Bias $\pm$ SE ($\hat{\sigma}$)	AMS	SSP	Bias $\pm$ SE ($\hat{\sigma}$)	AMS	SSP
PMLE	λ_1	0.163 ± 0.058	7.76	1.00	0.153 ± 0.052	9.80	1.00
	λ_2	0.259 ± 0.061	3.05	0.99	0.248 ± 0.052	5.21	1.00
	λ_3	0.372 ± 0.065	2.00	0.30	0.376 ± 0.054	3.82	1.00
BC	λ_1	0.047 ± 0.059	12.25	1.00	0.002 ± 0.054	15.65	1.00
	λ_2	0.162 ± 0.064	3.18	0.99	0.070 ± 0.054	5.82	1.00
	λ_3	0.339 ± 0.065	2.13	0.38	0.191 ± 0.058	4.25	1.00
Scaled Lasso	λ_1	-0.006 ± 0.059	14.97	1.00	-0.051 ± 0.057	18.78	1.00
	λ_2	0.130 ± 0.065	3.23	0.99	0.041 ± 0.056	6.05	1.00
	λ_3	0.320 ± 0.069	2.13	0.38	0.126 ± 0.061	4.46	1.00
Scaled MC+	λ_1	-0.132 ± 0.080	14.31	1.00	-0.075 ± 0.062	13.96	1.00
	λ_2	-0.008 ± 0.060	3.15	1.00	0.011 ± 0.056	3.25	1.00
	λ_3	0.150 ± 0.137	2.51	0.64	0.067 ± 0.061	2.98	0.96
Scaled SCAD	λ_1	-0.061 ± 0.057	14.13	1.00	-0.047 ± 0.059	14.36	1.00
	λ_2	0.083 ± 0.094	3.14	1.00	0.034 ± 0.057	3.94	1.00
	λ_3	0.319 ± 0.072	2.21	0.42	0.119 ± 0.061	3.94	1.00
N-LASSO		-0.526 ± 0.195	36.540	1.000	-0.457 ± 0.200	29.610	1.000
RCV-SIS		0.018 ± 0.137	50.000	0.870	-0.005 ± 0.135	50.000	1.000
RCV-ISIS		0.051 ± 0.168	30.845	0.710	0.022 ± 0.118	28.990	0.760
RCV-LASSO		-0.001 ± 0.129	31.070	0.920	-0.030 ± 0.113	26.520	0.990
P-SCAD		-0.137 ± 0.114	30.020	1.000	-0.116 ± 0.169	29.890	0.990
CV-SCAD		0.064 ± 0.121	30.020	1.000	0.092 ± 0.132	29.890	0.990
P-LASSO		-0.081 ± 0.207	36.540	1.000	-0.086 ± 0.153	29.610	1.000
CV-LASSO		0.141 ± 0.111	36.540	1.000	0.082 ± 0.104	29.610	1.000

and a plug-in method (P) with degrees-of-freedom correction. In addition to the bias and the standard error for the estimation of the noise level σ , we report the average model size (AMS) $|\hat{S}|$ and the relative frequency of sure screening (SSP) $\hat{S} \supset S$ as in Fan, Guo and Hao (2010), where $\hat{S} = \{j : \hat{\beta}_j \neq 0\}$ is the selected model.

We observe that the scaled Lasso at λ_1 and the scaled MC+ at λ_2 clearly outperform the cross-validation methods for the estimation of σ , especially for the standard deviation. The results of the average model size and sure screening probability show that cross-validation methods select about 30 variables when the true model size is 3. Meanwhile, the scaled estimators choose correct models at the universal penalty level $\{(2/n) \log p\}^{1/2}$ with large probabilities.

4.2. Real data example. In this subsection, we demonstrate the performance of the scaled Lasso and MC+ with a data set reported in Scheetz et al. (2006).

This data set contains 18976 probes for 120 rats. Our goal is to find probes that are related to that of gene TRIM32, which has been found to cause Bardet-Biedl syndrome, a genetically heterogeneous disease of multiple organ systems including

Table 3: The probe sets identified by Lasso with cross validation, the scaled Lasso and MC+ at $\lambda_2 = \{2(\log p)/n\}^{1/2}$. #cov is the number of covariates being considered. The superscript Δ indicates probes not included in the smaller set of 200 probes with large correlation. * indicates the covariates with coefficients larger than $\lambda/2$.

Probe ID	C-V Lasso		Scaled Lasso		Scaled MC+	
#cov	200	3000	200	3000	200	3000
1369353.at	-0.00912*	-0.00713	-0.0073	-0.00403		
1370052.at Δ		0.00365				
1370429.at	-0.00322		-0.00878	-0.00936	-0.01637*	
1371242.at	-0.00666*					
1374106.at	0.00888*	0.01058	0.00745	0.00701	0.00847	0.01002
1374131.at	0.00407	0.0008				
1375585.at Δ				0.00058		
1384204.at			0.0007			
1387060.at Δ		0.0035				
1388538.at Δ		0.00142				
1389584.at	0.01716*	0.02539*	0.01997*	0.02118*	0.04575*	0.05049*
1393979.at	-0.00181		-0.0004			
1379079.at Δ		-0.00143				
1379495.at		0.00484	0.00171	0.001		
1379971.at	0.01356*	0.0131*	0.01125	0.00952		
1380033.at	0.00869*		0.00297		0.00675	
1380070.at Δ		0.00019				
1381787.at	-0.00205		-0.00211			
1382452.at Δ		0.01293*		0.00163		0.01291
1382835.at	0.01264*	0.00579	0.00415			
1383110.at	0.00903*	0.01999*	0.01497*	0.01669*	0.01580*	0.02301*
1383522.at	0.00303					
1383673.at	0.00554*	0.00612	0.00608	0.00647		
1383749.at	-0.01386*	-0.01085*	-0.01102	-0.00807	-0.00274	-0.00111
1383996.at	0.02501*	0.01782*	0.01888*	0.01552*	0.02507*	0.01919*
1385687.at Δ		-0.00099				
1386683.at				0.0029		
1390788.a.at	0.00092					
1392692.at Δ		0.00174				
1393382.at	0.00243					
1393684.at	0.00159					
1395076.at Δ				0.00023		
1397489.at Δ		0.00333				
Model size	19	20	15	14	7	6
$\lambda = \sigma\lambda_0$	0.0103	0.0163	0.0243	0.0315	0.0244	0.0304

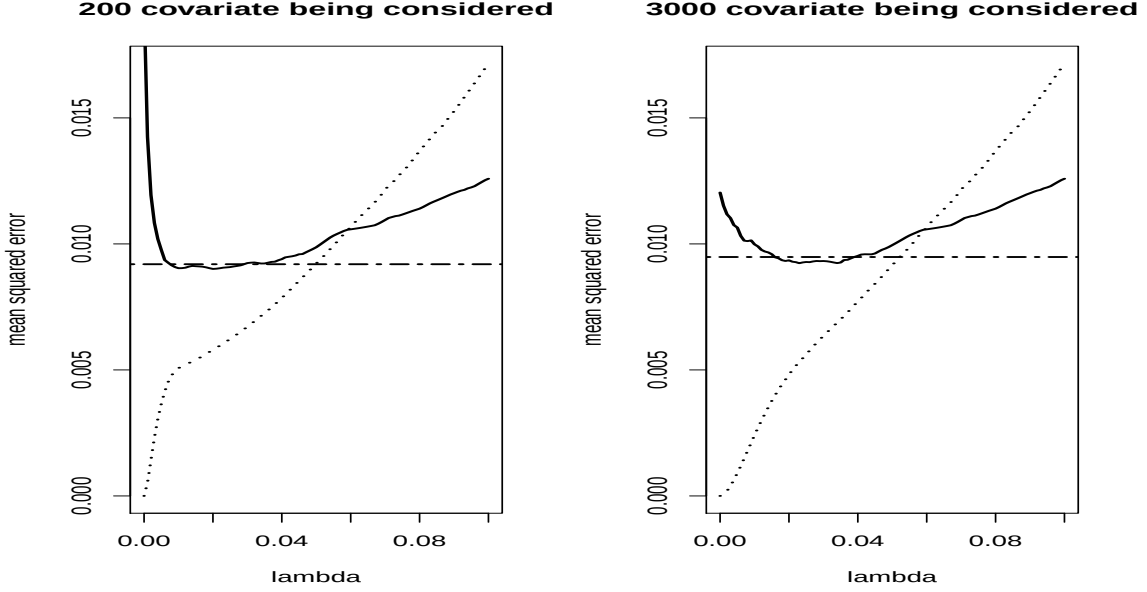


Figure 2: MSE against penalty level λ . Solid line: testing error for fixed λ ; dotted line: training error for fixed λ ; dashed line: testing error for the scaled Lasso.

the retina. We consider linear regression with the probe from TRIM32, 1389163_at, as the response variable. As in Huang, Ma and Zhang (2008), we focus on 3000 probes with the largest variances among the 18975 covariates and consider two approaches. The first is to regress on these $p = 3000$ probes. The second is to regress on the 200 probes among the 3000 with the largest marginal correlation coefficients with TRIM32. For the cross-validation Lasso, we randomly partition the data 1000 times, each with a training set of size 80 and a validation set of size 40. For each partition, the optimal penalty level λ is selected by minimizing the prediction mean squared error in the validation set. Then we compute the Lasso estimator with all 120 observations at the penalty level equal to the median of the optimal penalty values for the 1000 random partitions. For the MC+, we set $\gamma = 2/(1 - \sigma_{0.95}) = 6.37$, where $\sigma_{0.95}$ is the 95% quantile of $|x'_k x_j|/n$. Table 3 shows the probe sets identified by the cross validation Lasso and the scaled Lasso and MC+ at universal penalty level $\lambda_2 = \{2(\log p)/n\}^{1/2}$. We observe that the scaled Lasso chooses a more sparse model than the cross-validation Lasso, but includes all significant covariates identified by the cross-validation Lasso (with coefficient greater than $\lambda/2$) except for one. The model selected by the scaled MC+ is even more sparse, but with most significant covariates.

We are also interested in the prediction performance. We compute six estimators in 200 replications as follows. In each replication, the dataset is split at random into a training set with 80 observations and a test set with 40 observations. The

prediction MSE is the mean squared error in the test set. The PMLE, BC, and scaled Lasso, MC+ and SCAD are computed based on the training set. For the cross-validation, the training set of 80 observations is further partitioned at random 200 times into two groups of sizes 60 and 20 respectively, and an optimal penalty level is obtained using the estimated loss in the smaller group for the Lasso estimator based on the larger group. This optimal penalty is then used for the Lasso with the entire training set. Thus, the cross-validation Lasso is also based on the training set with 80 observations. We present the medians of the prediction MSE in the 200 replications in Figure 2 and Table 4. Figure 2 demonstrates that the scaled Lasso with λ_0 chosen as $\lambda_2 = \{2(\log p)/n\}^{1/2}$ performs almost as well as the Lasso with the optimal fixed λ in prediction. Table 4 demonstrates that the scaled Lasso has comparable prediction performance as cross validation. Again, Table 4 suggests that cross validation tends to choose larger models.

Table 4: Prediction performance.

Method		#cov = 200			#cov = 3000		
		P-MSE	$ \hat{\beta} _0$	corr	P-MSE	$ \hat{\beta} _0$	corr
PMLE	λ_1	0.0091	12	0.691	0.0093	12	0.656
	λ_2	0.0094	9	0.656	0.0101	8	0.618
	λ_3	0.0106	6	0.597	0.0117	3	0.497
BC	λ_1	0.0090	12	0.694	0.0093	15	0.655
	λ_2	0.0093	10	0.671	0.0096	9	0.631
	λ_3	0.0103	7	0.613	0.0113	4	0.508
Scaled Lasso	λ_1	0.0090	13	0.694	0.0092	16	0.657
	λ_2	0.0092	11	0.673	0.0095	10	0.638
	λ_3	0.0098	8	0.622	0.0109	5	0.517
Scaled MC+	λ_1	0.0100	6	0.671	0.0105	8	0.647
	λ_2	0.0101	5	0.650	0.0102	4	0.630
	λ_3	0.0105	3	0.616	0.0112	2	0.572
Scaled SCAD	λ_1	0.0096	11	0.691	0.0099	15	0.672
	λ_2	0.0093	11	0.697	0.0092	10	0.675
	λ_3	0.0095	8	0.664	0.0109	5	0.617
C-V Lasso		0.0091	14	0.698	0.0097	25	0.653

5. Discussion. We have proposed the scaled Lasso and MC+ with the iterative algorithm (3) for the joint estimation of the regression coefficients β and noise level σ . Sun and Zhang (2010) demonstrated the advantages of these algorithms over the PMLE of Städler, Bühlmann and van de Geer (2010) with a small simulation experiment. Our more extensive numerical experiments further support the earlier results and demonstrate the advantage of the scaled methods over cross-validation for the choice of the penalty level. For the scaled Lasso, we have established oracle

inequalities for the estimation of both β and σ via the convexity of the joint loss (4) by Huber (1981) and Antoniadis (2010). These oracle inequalities provide the consistency and the asymptotic normality for the estimation of σ , including the case of $p > n$.

Although theoretical results for the scaled Lasso is more complete, quite a few existing theoretical and simulation studies (Fan and Peng, 2004; Zhang, 2010; Mazumder, Friedman and Hastie, 2009) have supported the use of concave penalized LSE (MC+ and SCAD). For variable selection, the Lasso requires a highly restrictive irreproducible condition on the design matrix which does not scale with the dimension $|S|$ of the oracle model, while concave penalized LSE requires conditions of much weaker form in comparison. Model selection consistency implies oracle properties in estimation and prediction in the sense of matching the performance of an oracle expert with the knowledge of the set of relevant variables $S = \{j : \beta_j^* \neq 0\}$. An important issue with the concave penalized LSE is the multiplicity of local minimizers of the penalized loss. On this issue, Zhang (2010) proved selection consistency and oracle properties of the MC+ estimator for the local minimum computed by the PLUS algorithm. Quite a few simulation studies have demonstrated superiority of the concave penalized LSE in settings where variable selection fails, as in some configurations in Example 2 of our simulation experiment. However, theoretical results for the estimation of β and prediction typically require somewhat stronger condition for the concave penalized MLE than the Lasso.

Our theoretical results for the scaled Lasso are derived from oracle inequalities for fixed penalty which sharpen and unify existing results. We use the rest of the section to discuss this aspect of our results. Define

$$\eta^*(\lambda, \xi) = \min_T 2^{-1} \left[\eta(\lambda, \xi, \beta^*, T) + \left\{ \eta^2(\lambda, \xi, \beta^*, T) - 16\lambda^2 |\beta_{T^c}^*|^2 \right\}^{1/2} \right].$$

as a sharper version of the $\eta(\lambda, \xi, \beta^*, T)$ in (7).

Theorem 3 *Let $\widehat{\beta}(\lambda)$ be the minimizer of (1) with $\rho(t) = t$. Let $\beta^* \in \mathbb{R}^p$ be a target vector and $\xi > 1$. Then, in the event $|X'(y - X\beta^*)|_\infty/n \leq \lambda(\xi - 1)/(\xi + 1)$,*

$$|X\widehat{\beta}(\lambda) - X\beta^*|_2^2/n \leq \min \{ \eta_*(\lambda, \xi), \eta^*(\lambda, \xi) \} \quad (21)$$

with $\eta_(\lambda, \xi)$ in (9). Moreover, in the same event and with the $\mu(\lambda, \xi)$ in (12),*

$$|\widehat{\beta}(\lambda) - \beta^*|_1 \leq \mu(\lambda, \xi). \quad (22)$$

Since $\eta_*(\lambda, \xi) = \min_{w, T} \eta(\lambda, \xi, w, T)$ over all $\{w, T\}$, the prediction oracle inequality (21) is sharper than the upper bounds of Koltchinskii, Tsybakov and Lounici (2010) where the minimum is taken over $w_{T^c} = 0$ or $T = \emptyset$ and somewhat cruder constants are used. Similarly, (21) is sharper than the upper bound $\eta(\lambda, \xi, \beta^*, S)$ of van de Geer and Bühlmann (2009). Here and in the rest of the section, $S = \{j : \beta_j^* \neq 0\}$.

Since $\nu = 0+$ is allowed in (12) for $T = S$, $\mu(\lambda, \xi) \leq \{(\xi + 1)/2\}\lambda|S|/\kappa^2(\xi, S)$. Thus, the ℓ_1 oracle inequality (22) is at least a constant factor of 2 sharper than the upper bound $2\xi\lambda|S|/\kappa^2(\xi, S)$ in van de Geer and Bühlmann (2009). For the general case allowing $\beta_T^* \neq 0$ as in (12), Zhang (2009) and Ye and Zhang (2010) have obtained oracle inequalities based on sparse eigenvalues and the SCIF (16) respectively.

Appendix. Here we prove Proposition 1 and Theorems 3, 1 and 2.

Proof of Proposition 1. (i) Since $\hat{\beta} = \hat{\beta}(\sigma\lambda_0)$ is a solution of (2) at $\lambda = \sigma\lambda_0$,

$$\left((\partial/\partial w)L_{\lambda_0}(w, \sigma) \Big|_{w=\hat{\beta}(\sigma\lambda_0)} \right)_j = 0, \quad \forall \hat{\beta}_j(\sigma\lambda_0) \neq 0.$$

Since $\{j : \hat{\beta}_j(\lambda)\}$ is unchanged in a neighborhood of $\sigma\lambda_0$, $\{(\partial/\partial\sigma)(\hat{\beta}(\sigma\lambda_0)/\sigma)\}_j = 0$ for $\hat{\beta}_j(\sigma\lambda_0) = 0$. Thus,

$$\frac{\partial}{\partial\sigma}L_{\lambda_0}(\hat{\beta}(\sigma\lambda_0), \sigma) = \frac{\partial}{\partial t}L_{\lambda_0}(\hat{\beta}(\sigma\lambda_0), t) \Big|_{t=\sigma} = \frac{1-a}{2} - \frac{|y - X\hat{\beta}(\sigma\lambda_0)|_2^2}{2n\sigma^2}.$$

(ii) The convergence of (3) follows from the convexity (Huber, 1981, page 179; Antoniadis, 2010). The scale invariance follows from $L_0(c\beta, c\sigma; X, cy) = cL_0(\beta, \sigma; X, y)$, where $L_0(\beta, \sigma; X, y)$ expresses the dependence of (4) on the data (X, y) . $\square$

Proof of Theorem 3. (i) Let $h = \hat{\beta} - \beta^*$. Since $\sigma^*z^* = |X'(y - X\beta^*)|_\infty/n \leq \lambda(\xi - 1)/(\xi + 1)$, the basic inequality (20) implies that, for any vector w and set T ,

$$(Xh)'(X\hat{\beta} - Xw)/n \leq \frac{2\xi\lambda}{\xi + 1}|(w - \hat{\beta})_T|_1 + 2\lambda|w_{T^c}|_1 - \frac{2\lambda}{\xi + 1}|(w - \hat{\beta})_{T^c}|_1.$$

Since $2(Xh)'(X\hat{\beta} - Xw) = |X\hat{\beta} - Xw|_2^2 + |Xh|_2^2 - |X\beta^* - Xw|_2^2$, we find

$$|X\hat{\beta} - Xw|_2^2/n + |Xh|_2^2/n + (2b/\xi)|(w - \hat{\beta})_{T^c}|_1 \leq 2c + 2b|(w - \hat{\beta})_T|_1. \quad (23)$$

where $b = 2\xi\lambda/(\xi + 1)$ and $c = |X\beta^* - Xw|_2^2/(2n) + 2\lambda|w_{T^c}|_1$. We want to prove

$$|Xh|_2^2/n \leq \eta(\lambda, \xi, w, T) = 2c + b^2/a \quad \text{with} \quad a = \kappa^2(\xi, T)/|T|.$$

It suffices to consider the case $|Xh|_2^2/n \geq 2c$, where $\hat{\beta} - w \in \mathcal{C}(\xi, T)$ by (23) and

$$a|(w - \hat{\beta})_T|_1^2 \leq |X\hat{\beta} - Xw|_2^2/n \quad (24)$$

and by (8). Let $x = |(w - \hat{\beta})_T|_1$, $y = |X\hat{\beta} - Xw|_2^2/n$, and $z = |Xh|_2^2/n$. It follows from (23) and (24) that (x, y, z) satisfies the inequalities $y + z \leq 2c + 2bx$ and $ax^2 \leq y$. For such (x, y, z) the maximum of z is attained when $z - 2c = 2bx - y = 2bx - ax^2 = b^2/a$, attained when $x = b/a$. This gives $z \leq 2c + b^2/a = \eta(\lambda, \xi, w, T)$.

For $w = \beta^*$, it suffices to consider the case $z = (y + z)/2 > c = 2\lambda|\beta_{T^c}^*|_1$, where the cone condition holds for $\hat{\beta} - w$. Now, (x, y, z) satisfies an additional constraint $y = z$, resulting in $ax^2 \leq z \leq c + bx$. The maximum of z , attained at $ax^2 = c + bx$, is

$$c + b(b + \sqrt{b^2 + 4ac})/(2a) = \{\eta(\lambda, \xi, \beta^*, T) + \sqrt{\eta^2(\lambda, \xi, \beta^*, T) - 4c^2}\}/2. \quad (25)$$

(ii) Let $0 < \nu < 1$ and $T \subset \{1, \dots, p\}$. It follows from (23) with $w = \beta^*$ that

$$(1 + \xi)|Xh|_2^2/n + 2\lambda|h_{T^c}|_1 \leq 2\lambda(\xi + 1)|\beta_{T^c}^*|_1 + 2\xi\lambda|h_T|_1.$$

It suffices to consider $\nu|h|_1 \geq (\xi + 1)|\beta_{T^c}^*|_1$. In this case

$$(1 + \xi)|Xh|_2^2/n + 2\lambda(1 - \nu)|h_{T^c}|_1 \leq 2\lambda(\xi + \nu)|h_T|_1.$$

Thus, $(1 - \nu)|h_{T^c}|_1 \leq (\xi + \nu)|h_T|_1$, or equivalently $h \in \mathcal{C}((\xi + \nu)/(1 - \nu), T)$. It follows from (8) that $|Xh|_2^2/n \geq |h_T|_1^2 \kappa^2((\xi + \nu)/(1 - \nu), T)/|T|$, so that

$$(1 + \xi)|h_T|_1^2 \kappa^2((\xi + \nu)/(1 - \nu), T)/|T| + 2(1 - \nu)\lambda|h_{T^c}|_1 \leq 2(\xi + \nu)\lambda|h_T|_1. \quad (26)$$

Let $x = |h_T|_1$ and $y = |h_{T^c}|_1$. Write (26) as $ax^2 + by \leq cx$. Subject to this inequality, the maximum of $x + y$ is $\max_{x \geq 0} \{x + (cx - ax^2)/b\}$. This maximum, attained at $2ax = b + c$, is $x(b + c)/(2b) = (b + c)^2/(4ab)$. Thus,

$$|h|_1 \leq \frac{\{2(\xi + 1)\lambda\}^2|T|}{4(1 + \xi)\kappa^2((\xi + \nu)/(1 - \nu), T)\{2(1 - \nu)\lambda\}} = \frac{(\xi + 1)\lambda|T|/(1 - \nu)}{2\kappa^2((\xi + \nu)/(1 - \nu), T)}.$$

This gives $|h|_1 \leq \mu(\lambda, \xi)$ for $\nu|h|_1 \geq (\xi + 1)|\beta_{T^c}^*|_1$. $\square$

Proof of Theorem 1. Consider $t \geq \sigma^*(1 - \tau_0)$ and the penalty level $\lambda = t\lambda_0$ for the Lasso. Since $z^*\sigma^* \leq \sigma^*(1 - \tau_0)\lambda_0(\xi - 1)/(\xi + 1) \leq \lambda(\xi - 1)/(\xi + 1)$ and $\sigma^* = |y - X\hat{\beta}(t\lambda_0)|_2/\sqrt{n}$, the Cauchy-Schwarz and (21) imply

$$||y - X\hat{\beta}(t\lambda_0)|_2/\sqrt{n} - \sigma^*| \leq |X\hat{\beta}(t\lambda_0) - X\beta^*|_2/\sqrt{n} \leq \eta_*^{1/2}(t\lambda_0, \xi).$$

Since $\eta_*^{1/2}(t\lambda_0, \xi) \leq \sigma^*\tau_0$ for $t < \sigma^*$, the derivative (6) of the profile loss satisfies

$$2t^2 \frac{\partial}{\partial t} L_{\lambda_0}(\hat{\beta}(t\lambda_0), t) = t^2 - |y - X\hat{\beta}(t\lambda_0)|_2^2/n \leq t^2 - (\sigma^*)^2(1 - \tau_0)^2 = 0$$

at $t = \sigma^*(1 - \tau_0)$. This implies $\hat{\sigma} \geq \sigma^*(1 - \tau_0)$ by the strict convexity of the profile loss. For $t > \sigma^*$, $\eta_*^{1/2}(t\lambda_0, \xi) \leq t\tau_0$ by (7) and (9), so that

$$t^2 - |y - X\hat{\beta}(t\lambda_0)|_2^2/n \geq t^2 - (\sigma^* + t\tau_0)^2 = 0$$

at $t = \sigma^*/(1 - \tau_0)_+$. This implies $\hat{\sigma} \leq \sigma^*/(1 - \tau_0)_+$ by convexity. It follows that

$$|X\hat{\beta} - X\beta^*|_2/\sqrt{n} \leq \eta_*^{1/2}(\hat{\sigma}\lambda_0, \xi) \leq \hat{\sigma}\tau_0 \leq \sigma^*\tau_0/(1 - \tau_0)_+.$$

Since $P_{\beta, \sigma}\{|X'(y - X\beta)/n|_\infty \leq \sigma\sqrt{(2/n)\log p}\} \rightarrow 1$, (11) follows from (10). $\square$

The proof of Theorem 2 requires the following lemma.

Lemma 1 *Let T_m have the t -distribution with m degrees of freedom. Then, there exists $\epsilon_m \rightarrow 0$ such that for all $t > 0$*

$$P\left\{T_m^2 > m(e^{2t^2/(m-1)} - 1)\right\} \leq \frac{1 + \epsilon_m}{\sqrt{\pi}t} e^{-t^2}. \quad (27)$$

Proof. Let $x = \sqrt{m(e^{2t^2/(m-1)} - 1)}$. Since T_m has the t -distribution,

$$\begin{aligned} P\left\{T_m^2 > x^2\right\} &= \frac{2\Gamma((m+1)/2)}{\Gamma(m/2)\sqrt{m\pi}} \int_x^\infty \left(1 + \frac{u^2}{m}\right)^{-(m+1)/2} du \\ &\leq \frac{2\Gamma((m+1)/2)}{x\Gamma(m/2)\sqrt{m\pi}} \int_x^\infty \left(1 + \frac{u^2}{m}\right)^{-(m+1)/2} u du \\ &= \frac{2\Gamma((m+1)/2)m}{x\Gamma(m/2)\sqrt{m\pi}(m-1)} \left(1 + \frac{x^2}{m}\right)^{-(m-1)/2}. \end{aligned}$$

Since $x \geq t\sqrt{2m/(m-1)}$,

$$P\left\{T_m^2 > x^2\right\} \leq \frac{\sqrt{2}\Gamma((m+1)/2)}{\Gamma(m/2)\sqrt{m-1}} \frac{e^{-t^2}}{t\sqrt{\pi}} = (1 + \epsilon_m) \frac{e^{-t^2}}{t\sqrt{\pi}},$$

where $\epsilon_m = \sqrt{2}\Gamma((m+1)/2)/\{\Gamma(m/2)\sqrt{m-1}\} - 1 \rightarrow 0$ as $m \rightarrow \infty$. $\square$

Proof of Theorem 2. (i) Consider $z^* \leq (1 - \tau_-)\lambda_0(\xi - 1)/(\xi + 1)$. Let $\lambda = t\lambda_0$ and $h = \widehat{\beta}(\lambda) - \beta^*$. Since $|X'(y - X\beta^*)/n|_\infty = z^*\sigma^*$, the KKT conditions (2) give

$$\begin{aligned} -(z^*\sigma^* + \lambda)|h|_1 &\leq (Xh)'(y - X\beta^* + y - X\widehat{\beta}(\lambda))/n \\ &= (\sigma^*)^2 - |y - X\widehat{\beta}(\lambda)|_2^2/n \\ &= (Xh)'\{2(y - X\beta^*) - Xh\}/n \leq 2z^*\sigma^*|h|_1. \end{aligned} \quad (28)$$

For $t \geq \sigma^*(1 - \tau_-)$, $z^*\sigma^* \leq t\lambda_0(\xi - 1)/(\xi + 1) = \lambda(\xi - 1)/(\xi + 1)$, so that (22) in Theorem 3 implies $|h|_1 \leq \mu(t\lambda_0, \xi)$. It follows that for $t = \sigma^*(1 - \tau_-)$,

$$\begin{aligned} t^2 - |y - X\widehat{\beta}(t\lambda_0)|_2^2/n &\leq t^2 - (\sigma^*)^2 + 2z^*\sigma^*\mu(t\lambda_0, \xi) \\ &\leq 2t(t - \sigma^*) + 2t\lambda_0(\xi - 1)(\xi + 1)^{-1}\mu(\sigma^*\lambda_0, \xi) = 0, \end{aligned}$$

due to $\tau_- = (\xi - 1)(\xi + 1)^{-1}\tau(\sigma^*) = (\xi - 1)(\xi + 1)^{-1}\lambda_0\mu(\sigma^*\lambda_0, \xi)/\sigma^*$. As in the proof of Theorem 1, we find $\widehat{\sigma}/\sigma^* \geq 1 - \tau_-$ by (6) and the strict convexity of (4).

To prove $\widehat{\sigma}/\sigma^* \leq 1 + \tau_+$, we set $\tau = \tau(\sigma^*)$ and consider $\tau < 1$. For $t > \sigma^*$, $\mu(t\lambda_0, \xi) \leq (t/\sigma^*)\mu(\sigma^*\lambda_0, \xi)$ by (12). Thus, since $(\xi - 1)/(\xi + 1) + 1 = 2\tau_+(1 - \tau)/\tau$ and $\tau_+ \leq (1 + \tau_+)\tau$, for $t/\sigma^* = 1 + \tau_+$ (28) and (22) imply

$$\begin{aligned} &t^2 - |y - X\widehat{\beta}(t\lambda_0)|_2^2/n \\ &\geq t^2 - (\sigma^*)^2 - (z^*\sigma^* + t\lambda_0)\mu(t\lambda_0, \xi) \\ &\geq (t + \sigma^*)\sigma^*\tau_+ - \{(\xi - 1)/(\xi + 1) + 1 + \tau_+\}t\lambda_0\mu(\sigma^*\lambda_0, \xi) \end{aligned}$$

$$\begin{aligned}
&= (\sigma^*)^2[(2 + \tau_+)\tau_+ - \{2\tau_+(1 - \tau)/\tau + \tau_+\}(1 + \tau_+)\tau] \\
&= (\sigma^*)^2\tau_+\{\tau(1 + \tau_+) - \tau_+\} > 0.
\end{aligned}$$

It follows that $\widehat{\sigma}/\sigma^* \leq 1 + \tau_+$ by convexity.

Since $1 - \tau_- \leq \widehat{\sigma}/\sigma^* \leq 1 + \tau_+$, $|\widehat{\beta}(\widehat{\sigma}\lambda_0) - \beta^*|_1 \leq \mu(\widehat{\sigma}\lambda_0, \xi) \leq \mu(\sigma^*\lambda_0, \xi)(1 + \tau_+)$. This completes the proof of (14).

(ii) Let $z_j = x'_j(y - X\beta^*)/(n\sigma^*)$, so that $z^* = \max_{j \leq p} |z_j|$. Under $P_{\beta^*, \sigma}$, $\epsilon^* = y - X\beta^*$ is a vector of iid normal variables with zero mean. Since $\sigma^* = |y - X\beta^*|/\sqrt{n}$, $z_j/\{(1 - z_j^2)/(n - 1)\}^{1/2}$ follows the t -distribution with $n - 1$ degrees of freedom. We use Lemma 1 with $m = n - 1$, $\epsilon_{n-1} = \delta_n$ and $t^2 = \log(p/\epsilon) > 2$ to obtain

$$P_{\beta^*, \sigma} \left\{ \frac{(n-1)z_j^2}{1 - z_j^2} > (n-1)(e^{2t^2/(n-2)} - 1) \right\} \leq \frac{1 + \delta_n}{\sqrt{\pi}t} e^{-t^2} = \frac{(1 + \delta_n)\epsilon/p}{\sqrt{\pi} \log(p/\epsilon)}. \quad (29)$$

Since $e^a - 1 \leq \sum_{k=1}^{\infty} a^k/2^{k-1} = a/(1 - a/2)$ for any $0 < a < 2$,

$$(n-1)(e^{2t^2/(n-2)} - 1) \leq \frac{(n-1) \cdot 2t^2/(n-2)}{1 - t^2/(n-2)} \leq \frac{(n-1) \cdot 2t^2/n}{1 - 2t^2/n}. \quad (30)$$

The combination of (29) and (30) yields

$$\begin{aligned}
P_{\beta^*, \sigma} \{ |z_j| > \sqrt{2 \log(p/\epsilon)/n} \} &= P_{\beta^*, \sigma} \left\{ \frac{(n-1)z_j^2}{1 - z_j^2} > \frac{(n-1) \cdot 2t^2/n}{1 - 2t^2/n} \right\} \\
&\leq P_{\beta^*, \sigma} \left\{ \frac{(n-1)z_j^2}{1 - z_j^2} > (n-1)(e^{2t^2/(n-2)} - 1) \right\} \\
&\leq (1 + \delta_n)(\epsilon/p)/\sqrt{\pi \log(p/\epsilon)}.
\end{aligned}$$

Since $\lambda_0 \geq \{(2/n) \log(p/\epsilon)\}^{1/2}(\xi + 1)/\{(\xi - 1)(1 - \tau_-)\}$, this bounds the tail probability of $z^* = \max_{j \leq p} |z_j|$ by the union bound. Since $n(\sigma^*/\sigma)^2 \sim \chi_n^2$, $\sqrt{n}(\sigma^*/\sigma - 1) \xrightarrow{D} N(0, 1/2)$, which then implies (15) by (14) under $\tau(\sigma) = o(n^{-1/2})$. $\square$

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